specguard report

`specguard/examples/arp4754b_appendix_e.py`

May 8, 2026

1 Encoded collections

`PSSA_REQUIREMENTS`

`E12-1` Airplane electrical power 1 independent from airplane electrical power 2

$$\neg(\text{lost\_elec\_input\_1} \wedge \text{lost\_elec\_input\_2})$$

`E12-2` NORMAL Mode braking function independent from ALTERNATE Mode braking function, for total loss failure condition

$$\neg(\text{failed\_normal\_braking} \wedge \text{failed\_alternate\_braking})$$

`E12-3` The NMVs and their control are independent from the SOV and its control such that no single failure results in uncommanded braking

$$(\text{single\_failure} \Rightarrow \neg\text{inadvertent\_braking\_takeoff})$$

`E13-1` The probability of BSCU failure resulting in loss of a valid braking command output to the NMV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_loss\_braking\_cmd} \leq 2 \times 10^{-4}$$

`E13-2` The probability of BSCU failure resulting in unannunciated erroneous braking command to the NMV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_erroneous\_cmd} \leq 2 \times 10^{-4}$$

`E13-3` The probability of BSCU failure resulting in the loss of command to open the SOV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_loss\_sov\_cmd} \leq 2 \times 10^{-4}$$

`E13-4` The probability of BSCU failure resulting in unintended closure of the S/ASV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_unintended\_sasv} \leq 2 \times 10^{-4}$$

`E13-5` The SOV and NMV commands shall be provided by the BSCU upon loss of either airplane electrical power input

$$(((\text{lost\_elec\_input\_1} \wedge \neg \text{lost\_elec\_input\_2}) \Rightarrow (\text{bscu\_provides\_sov\_cmd} \wedge \text{bscu\_provides\_nmv\_cmd})) \wedge ((\text{lost\_elec\_input\_2} \wedge \neg \text{lost\_elec\_input\_1}) \Rightarrow (\text{bscu\_provides\_sov\_cmd} \wedge \text{bscu\_provides\_nmv\_cmd})))$$

`E13-6` When "HYD 1 Enable" output is enabled, then "Alt/Emer Ctrl" output shall be disabled

$$(\text{hyd1\_enable\_on} \Rightarrow \neg \text{alt\_emer\_ctrl\_on})$$

`E13-7` When "HYD 1 Enable" output is disabled, then "Alt/Emer Ctrl" output shall be enabled

$$(\neg \text{hyd1\_enable\_on} \Rightarrow \text{alt\_emer\_ctrl\_on})$$

`E13-8` No single failure shall cause erroneous NMV command and inhibit the SOV function

$$\neg(\text{erroneous\_nmv\_cmd} \wedge \text{sov\_function\_inhibited})$$

`E13-9` The wheel brake command function of the BSCU shall be developed to FDAL A

$$\text{bscu\_cmd\_fdal\_a}$$

`E14-1` The probability of 'Loss of Normal Braking System Hydraulic Equipment' will be less than 3.3E-05 per flight

$$\text{p\_loss\_normal\_hyd\_equip} < 3.3 \times 10^{-5}$$

`E14-2` The probability of 'Loss of Alternate Braking System Hydraulic Equipment' will be less than 3.3E-05 per flight

$$\text{p\_loss\_alt\_hyd\_equip} < 3.3 \times 10^{-5}$$

`E14-3` The probability of 7 or more wheel speed sensors erroneous or inoperative will be less than 1.0E-07 per flight

$$\text{p\_seven\_or\_more\_sensors\_per\_flight} < 1 \times 10^{-7}$$

`E14-4` The probability of loss of an airplane electrical power bus will be less than 1.0E-04 per flight

$$\text{p\_loss\_elec\_bus\_per\_flight} < 1 \times 10^{-4}$$

`E14-5` The probability of loss of a Left brake pedal position input will be less than 1.0E-04 per flight

$$\text{p\_loss\_pedal\_per\_flight} < 1 \times 10^{-4}$$

`E14-6` Airplane electrical power bus 1 is independent from airplane electrical power bus 2

$$\neg(\text{lost\_elec\_input\_1} \wedge \text{lost\_elec\_input\_2})$$

`E14-7` HYD 1 hydraulic system is independent from HYD 2 hydraulic system

$$\neg(\text{failed\_hyd\_1} \wedge \text{failed\_hyd\_2})$$

ALT_PSSA_REQUIREMENTS

`E28-WBS-ASMP-1` The probability of 'Loss of Normal Braking System Hydraulic Equipment' will be less than 3.3E-05 per flight

$$\text{p\_loss\_normal\_hyd\_equip} < 3.3 \times 10^{-5}$$

`E28-WBS-ASMP-2` The probability of 'Loss of Alternate Braking System Hydraulic Equipment' will be less than 3.3E-05 per flight

$$\text{p\_loss\_alt\_hyd\_equip} < 3.3 \times 10^{-5}$$

`E28-WBS-ASMP-3` The probability of 7 or more wheel speed sensors erroneous or inoperative will be less than 1.0E-07 per flight

$$\text{p\_seven\_or\_more\_sensors\_per\_flight} < 1 \times 10^{-7}$$

`E28-WBS-ASMP-4` The failure rate for loss of an airplane electrical power bus will be less than 1.0E-04 per hour of flight

$$\text{lambda\_loss\_elec\_bus\_per\_hour} < 1 \times 10^{-4}$$

`E28-WBS-ASMP-5` The failure rate for loss of a left brake pedal position input will be less than 1.0E-04 per hour of flight

$$\text{lambda\_loss\_pedal\_per\_hour} < 1 \times 10^{-4}$$

`E28-WBS-ASMP-6` Airplane electrical power bus 1 is independent from airplane electrical power bus 2

$$\neg(\text{lost\_elec\_input\_1} \land \text{lost\_elec\_input\_2})$$

`E28-WBS-ASMP-7` HYD 1 hydraulic system is independent from HYD 2 hydraulic system

$$\neg(\text{failed\_hyd\_1} \land \text{failed\_hyd\_2})$$

SPEC_REQUIREMENTS

`S18-WBS-R-0020` The Wheel Brake System shall have a means to decelerate the wheels on the ground

$$\text{wbs\_has\_decelerate\_means}$$

`S18-WBS-R-0021` The Wheel Brake System shall be capable of decelerating the S18 airplane to a complete stop/to taxi speed in 2000 feet when wheel brakes, high lift speed brakes and reverse thrust are available, including when at maximum landing weight

$$\text{stopping\_distance\_ft} \leq 2000$$

`S18-WBS-R-0041` The Wheel Brake System shall provide directional control on ground by a differential braking function

$$\text{wbs\_differential\_braking}$$

`S18-WBS-R-0042` The Wheel Brake System shall provide a parking brake to prevent airplane motion on the ground

wbs_parking_brake

`S18-WBS-R-0043` The Wheel Brake System shall provide autobraking capability during landing and RTO

wbs_autobraking

`S18-WBS-R-0044` The Wheel Brake System shall provide anti-skid braking

wbs_antiskid

`S18-WBS-R-0045` The Wheel Brake System shall provide hydraulic brake control

wbs_hyd_brake_control

`S18-WBS-R-0046` The Wheel Brake System shall override the autobrake when the commanded by the flight crew

wbs_overrides_autobrake_on_crew

`S18-WBS-R-0047` The Wheel Brake System shall meet safety requirements while operating in an average atmospheric radiation environment per the IEC 62396 standard with an altitude of 40000 feet and a latitude of 45 degrees

wbs_radiation_qualified

`S18-WBS-R-0049` The Wheel Brake System shall be controlled and monitored by a computer system called Brake System Control Unit

wbs_controlled_by_bscu

`S18-WBS-R-0050` Each wheel brake shall have separate hydraulic power supply lines

each_wheel_separate_hyd_lines

`S18-WBS-R-0052` Each hydraulic Wheel Brake System circuit shall have meter valves, anti-skid valves and hydraulic fuses

each_circuit_valves_fuses

`S18-WBS-R-0055` Anti-skid system shall be capable to prevent skidding of the tires by reducing the pressure applied to the brakes

antiskid_prevents_skidding

`S18-WBS-R-0062` The Wheel Brake System shall include an emergency accumulator to supply hydraulic power to the wheel brakes

has_emergency_accumulator

`S18-WBS-R-0065` The emergency accumulator shall provide a minimum of 1800 psi

$$\text{accumulator\_pressure\_psi} \geq 1800$$

`S18-WBS-R-0066` The Alternate/Emergency Brake System hydraulic equipment and piping shall be installed aft of the engine 1 UERF trajectory envelope

$$\text{alt\_emer\_aft\_of\_uerf}$$

`S18-WBS-R-0100` The Wheel Brake System decelerate the wheels on the ground function shall be developed as FDAL A

$$\text{wbs\_decelerate\_fdal\_a}$$

`S18-WBS-R-0120` No single failure or event shall cause the complete loss of hydraulic power for both the WBS Normal and Alternate/Emergency Brake Systems

$$\neg(\text{failed\_normal\_braking} \wedge \text{failed\_alternate\_braking})$$

`S18-WBS-R-0130` No single failure or event shall cause the complete loss of electrical power for both the WBS Normal and Alternate/Emergency Brake Systems

$$\neg(\text{lost\_elec\_input\_1} \wedge \text{lost\_elec\_input\_2})$$

`S18-WBS-R-0150` Complete loss of decelerate the wheels on the ground function shall be less probable than 1.0E-07 for a landing

$$\text{p\_total\_loss\_decelerate} < 1 \times 10^{-7}$$

`S18-WBS-R-0200` Two redundant control lanes shall be provided between the EBU and each of the two Alternate/Emergency Meter Valves

$$\text{ebu\_two\_redundant\_lanes}$$

`S18-WBS-R-0326` No single failure shall result in inadvertent wheel braking of all wheels during takeoff roll

$$(\text{single\_failure} \Rightarrow \neg\text{inadvertent\_braking\_takeoff})$$

`S18-WBS-R-0508` The Wheel Brake System shall have at least two independent hydraulic pressure sources

$$\text{wbs\_two\_independent\_hyd\_sources}$$

`S18-WBS-R-0509` The Wheel Brake System shall have dual BSCU command functions

$$\text{wbs\_dual\_bscu\_cmd}$$

`S18-WBS-R-0510` The rudder and nose wheel steering functions shall not be implemented by the Wheel Brake System

$$\text{wbs\_no\_rudder\_or\_nws}$$

`S18-WBS-R-0631` The Wheel Brake System shall indicate individual brake temperature

$$\text{wbs\_indicates\_brake\_temp}$$

`S18-WBS-R-0632` The Wheel Brake System shall control individual brake pressure

$$\text{wbs\_individual\_brake\_pressure}$$

`S18-WBS-ICD-1002` The brake actuators shall respond to a BSCU command within 5 ms of receiving it

$$\text{actuator\_response\_time\_ms} \leq 5$$

`S18-WBS-R-1613` Hydraulic pressure shall be controlled for weight, autobrake mode, ground speed, wheel rotation, brake temperature and deceleration rate in accordance with graphs given in S18 brake force analysis AAS18-XXX

$$\text{hyd\_pressure\_per\_analysis}$$

`S18-WBS-R-2971` Wheel Brake System shall have one NORMAL operating mode

$$\text{wbs\_one\_normal\_mode}$$

`S18-WBS-R-2972` Wheel Brake System shall have one ALTERNATE operating mode

$$\text{wbs\_one\_alternate\_mode}$$

`S18-WBS-R-2973` Operation of the Alternate/Emergency Brake System shall be precluded when the Normal Brake System is in use

$$(\text{normal\_mode\_active} \Rightarrow \neg\text{alternate\_mode\_active})$$

`S18-WBS-R-2974` An emergency accumulator shall provide hydraulic power for EMERGENCY Wheel Brake System operating mode

$$\text{accumulator\_powers\_emergency}$$

`S18-WBS-R-2975` The emergency accumulator shall be attached to the HYD 2 hydraulic line between the Selector Valve and the Alternate/Emergency Meter Valve

$$\text{accumulator\_attached\_to\_hyd2}$$

`S18-WBS-R-2986` The wheel brake command function of the BSCU shall be developed as FDAL A

$$\text{bscu\_cmd\_fdal\_a}$$

`S18-WBS-R-2997` The BSCU shall have two independent command channels

$$\text{bscu\_two\_independent\_channels}$$

`S18-WBS-R-2999` The BSCU shall have two independent electrical power sources

$$\text{bscu\_two\_independent\_power}$$

`S18-WBS-R-3011` Brake pedals shall be independent from the rudder control

$$\text{brake\_pedals\_indep\_rudder}$$

`S18-WBS-R-3213` Normal Brake System shall be powered from the airplane's HYD 1 hydraulic system

$$(\text{failed\_hyd\_1} \Rightarrow \text{failed\_normal\_braking})$$

`S18-WBS-R-3224` Alternate/Emergency Brake System shall be powered from the airplane's HYD 2 hydraulic system

$$(\text{failed\_hyd\_2} \Rightarrow \text{failed\_alternate\_braking})$$

`S18-WBS-R-3245` ALTERNATE Mode shall be automatically selected when the NORMAL Mode fails

$$(\text{normal\_mode\_failed} \Rightarrow \text{alternate\_mode\_active})$$

`S18-WBS-R-6104` The probability of BSCU failure resulting in loss of a valid braking command output to the NMV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_loss\_braking\_cmd} \leq 2 \times 10^{-4}$$

`S18-WBS-R-6105` The probability of BSCU failure resulting in unannunciated erroneous braking command to the NMV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_erroneous\_cmd} \leq 2 \times 10^{-4}$$

`S18-WBS-R-6106` The probability of BSCU failure resulting in the loss of command to open the SOV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_loss\_sov\_cmd} \leq 2 \times 10^{-4}$$

`S18-WBS-R-6107` The probability of BSCU failure resulting in unintended closure of the S/ASV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_unintended\_sasv} \leq 2 \times 10^{-4}$$

`S18-WBS-R-6108` The SOV and NMV commands shall be provided by the BSCU upon loss of either airplane electrical power input

$$(((\text{lost\_elec\_input\_1} \land \neg \text{lost\_elec\_input\_2}) \Rightarrow (\text{bscu\_provides\_sov\_cmd} \land \text{bscu\_provides\_nmv\_cmd})) \\ \land ((\text{lost\_elec\_input\_2} \land \neg \text{lost\_elec\_input\_1}) \Rightarrow (\text{bscu\_provides\_sov\_cmd} \land \text{bscu\_provides\_nmv\_cmd})))$$

`S18-WBS-R-6109` When "HYD 1 Enable" output is enabled, then "Alt/Emer Ctrl" output shall be disabled

$$(\text{hyd1\_enable\_on} \Rightarrow \neg \text{alt\_emer\_ctrl\_on})$$

`S18-WBS-R-6110` No single failure shall cause erroneous NMV command and inhibit the SOV function

$$\neg(\text{erroneous\_nmv\_cmd} \wedge \text{sov\_function\_inhibited})$$

`S18-WBS-R-6111` The probability of 'Loss of Normal Brake System Hydraulic Components' shall be less than 3.3E-05 per flight

$$\text{p\_loss\_normal\_hyd\_equip} < 3.3 \times 10^{-5}$$

`S18-WBS-R-6112` The probability of 'Loss of Alternate/Emergency Brake System Hydraulic Components' shall be less than 3.3E-05 per flight

$$\text{p\_loss\_alt\_hyd\_equip} < 3.3 \times 10^{-5}$$

`BSCU_PSSA_REQUIREMENTS`

`BSCU-001` Channel #1 shall have physical independence from Channel #2

$$\text{channel1\_phys\_indep\_channel2}$$

`BSCU-002` Within each channel, the command function shall have physical independence from the monitor function

$$\text{cmd\_phys\_indep\_monitor}$$

`BSCU-003` The COMX software and MONX software shall be developed using different software requirement specifications

$$\text{comx\_monx\_diff\_specs}$$

`BSCU-004` Within each channel, a power supply monitor shall shutoff the power supply when any voltage is detected to be out of specification

$$(\text{voltage\_out\_of\_spec} \Rightarrow \text{power\_supply\_shutoff})$$

`BSCU-005` Within each channel, the operational state of the power supply shall have no effect on the operational state of the power supply monitor

$$\text{power\_supply\_state\_indep\_monitor}$$

`BSCU-006` Upon Channel #1 indicating an 'invalid' state, Channel #2 shall command the 'Normal Control' output

$$(\text{channel1\_invalid} \Rightarrow \text{channel2\_cmd\_normal\_ctrl})$$

`BSCU-007` Upon both Channel #1 and Channel #2 indicating 'invalid' states, the 'HYD 1 Enable' output shall be disabled

$$((\text{channel1\_invalid} \wedge \text{channel2\_invalid}) \Rightarrow \neg \text{hyd1\_enable\_on})$$

`BSCU-008` Upon either Channel #1 or Channel #2 indicating a validity error, a WBS annunciation shall be output from the BSCU

$$((\text{channel1\_invalid} \lor \text{channel2\_invalid}) \Rightarrow \text{wbs\_annunciation\_out})$$

`BSCU-009` At power-up the BSCU shall perform a self-test of Channel #2 including the ability to switch control from Channel #1 to Channel #2

$$\text{power\_up\_self\_tests\_ch2}$$

`BSCU-010` At power-up each channel of the BSCU shall implement a test of its power supply monitor

$$\text{power\_up\_tests\_psm}$$

`BSCU-011` If the power-up test fails, a WBS annunciation shall be output from the BSCU

$$(\text{power\_up\_test\_failed} \Rightarrow \text{wbs\_annunciation\_out})$$

`BSCU_ASSUMPTIONS_TO_WBS`

`E23-1` When 'HYD 1 Enable' is disabled, the BSCU brake control outputs are not used

$$(\neg \text{hyd1\_enable\_on} \Rightarrow \neg \text{bscu\_brake\_outputs\_in\_use})$$

`E23-2` A BSCU maintenance action will be initiated, if the WBS Status indicates a failure

$$(\text{wbs\_status\_failure} \Rightarrow \text{bscu\_maintenance\_initiated})$$

`BSCU_SPEC_REQUIREMENTS`

`S18-BSCU-R-0001` The wheel brake command function of the BSCU shall be developed as FDAL A

$$\text{bscu\_cmd\_fdal\_a}$$

`S18-BSCU-R-0002` The probability of BSCU failure resulting in loss of a valid braking command output to the NMV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_loss\_braking\_cmd} \leq 2 \times 10^{-4}$$

`S18-BSCU-R-0003` The probability of BSCU failure resulting in unannunciated erroneous braking command to the NMV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_erroneous\_cmd} \leq 2 \times 10^{-4}$$

`S18-BSCU-R-0004` The probability of BSCU failure resulting in the loss of command to open the SOV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_loss\_sov\_cmd} \leq 2 \times 10^{-4}$$

`S18-BSCU-R-0005` The probability of BSCU failure resulting in unintended closure of the S/ASV shall not exceed 2.0E-04 per flight

$$\text{p\_bscu\_unintended\_sasv} \leq 2 \times 10^{-4}$$

`S18-BSCU-R-0006` The SOV and NMV commands shall be provided by the BSCU upon loss of either airplane electrical power input

$$(((\text{lost\_elec\_input\_1} \wedge \neg \text{lost\_elec\_input\_2}) \Rightarrow (\text{bscu\_provides\_sov\_cmd} \wedge \text{bscu\_provides\_nmv\_cmd})) \\ \wedge ((\text{lost\_elec\_input\_2} \wedge \neg \text{lost\_elec\_input\_1}) \Rightarrow (\text{bscu\_provides\_sov\_cmd} \wedge \text{bscu\_provides\_nmv\_cmd})))$$

`S18-BSCU-R-0007` When "HYD 1 Enable" output is enabled, then "Alt/Emer Ctrl" output shall be disabled

$$(\text{hyd1\_enable\_on} \Rightarrow \neg \text{alt\_emer\_ctrl\_on})$$

`S18-BSCU-R-0008` No single failure shall cause erroneous NMV command and inhibit the SOV function

$$\neg(\text{erroneous\_nmv\_cmd} \wedge \text{sov\_function\_inhibited})$$

`S18-BSCU-R-0009` When 'HYD 1 Enable' is disabled, the BSCU brake control outputs shall be disabled

$$(\neg \text{hyd1\_enable\_on} \Rightarrow \neg \text{bscu\_brake\_outputs\_in\_use})$$

`S18-BSCU-R-0010` BSCU shall calculate the required hydraulic pressure as a function of weight, autobrake mode, ground speed, wheel rotation, brake temperature and deceleration rate in accordance with graphs given in S18 brake force analysis AAS18-XXX

$$\text{hyd\_pressure\_per\_analysis}$$

`S18-BSCU-R-0011` BSCU shall control the meter valves

$$\text{bscu\_controls\_meter\_valves}$$

`S18-BSCU-R-0012` BSCU shall control the anti-skid valves

$$\text{bscu\_controls\_antiskid\_valves}$$

`S18-BSCU-R-0013` BSCU shall control the shut off valves

$$\text{bscu\_controls\_shutoff\_valves}$$

`S18-BSCU-R-0027` Each BSCU shall have two command channels

$$\text{bscu\_each\_two\_command\_channels}$$

`S18-BSCU-R-0051` Each BSCU channel shall have a monitoring unit

$$\text{bscu\_each\_channel\_has\_monitoring}$$

`S18-BSCU-R-0075` Each BSCU channel shall have a control unit

$$\text{bscu\_each\_channel\_has\_control}$$

`S18-BSCU-R-0099` Each BSCU channel shall have independent power

$$\text{bscu\_each\_channel\_indep\_power}$$

`S18-BSCU-R-0123` Each BSCU channel shall have independent pedal input to both control and monitoring units

$$\text{bscu\_each\_channel\_indep\_pedal}$$

`S18-BSCU-R-0124` The command channels of each BSCU shall be independent of each other

$$\text{bscu\_command\_channels\_indep}$$

`S18-BSCU-R-0126` The BSCU hardware shall be developed to IDAL A

$$\text{bscu\_hw\_idal\_a}$$

`S18-BSCU-R-0127` The BSCU software shall be developed to IDAL B

$$\text{bscu\_sw\_idal\_b}$$

`S18-BSCU-R-0128` BSCU shall meet safety requirements while operating in an average atmospheric radiation environment per the IEC 62396 standard with an altitude of 40000 feet and a latitude of 45 degrees

$$\text{bscu\_radiation\_qualified}$$

`S18-BSCU-R-0201` Channel #1 shall have physical independence from Channel #2

$$\text{channel1\_phys\_indep\_channel2}$$

`S18-BSCU-R-0202` Within each channel, the command function shall have physical independence from the monitor function

$$\text{cmd\_phys\_indep\_monitor}$$

`S18-BSCU-R-0203` The COMX software and MONX software shall be developed using different software requirement specifications

$$\text{comx\_monx\_diff\_specs}$$

`S18-BSCU-R-0204` Within each channel, a power supply monitor shall shutoff the power supply when any voltage is detected to be out of specification

$$(\text{voltage\_out\_of\_spec} \Rightarrow \text{power\_supply\_shutoff})$$

`S18-BSCU-R-0205` Within each channel, the operational state of the power supply shall have no effect on the operational state of the power supply monitor

$$\text{power\_supply\_state\_indep\_monitor}$$

`S18-BSCU-R-0206` Upon Channel #1 indicating an 'invalid' state, Channel #2 shall command the 'Normal Control' output

$$(\text{channel1\_invalid} \Rightarrow \text{channel2\_cmd\_normal\_ctrl})$$

`S18-BSCU-R-0207` Upon both Channel #1 and Channel #2 indicating 'invalid' states, the 'HYD 1 Enable' output shall be disabled

$$((\text{channel1\_invalid} \wedge \text{channel2\_invalid}) \Rightarrow \neg \text{hyd1\_enable\_on})$$

`S18-BSCU-R-0208` Upon either Channel #1 or Channel #2 indicating a validity error, a WBS annunciation shall be output from the BSCU

$$((\text{channel1\_invalid} \vee \text{channel2\_invalid}) \Rightarrow \text{wbs\_annunciation\_out})$$

`S18-BSCU-R-0209` At power-up, the BSCU shall perform a self-test of Channel #2 including the ability to switch control from Channel #1 to Channel #2

$$\text{power\_up\_self\_tests\_ch2}$$

`S18-BSCU-R-0210` At power-up, each channel of the BSCU shall implement a test of its power supply monitor

$$\text{power\_up\_tests\_psm}$$

`S18-BSCU-R-0211` If the power-up test fails, a WBS annunciation shall be output from the BSCU

$$(\text{power\_up\_test\_failed} \Rightarrow \text{wbs\_annunciation\_out})$$

`AIRPLANE_REQUIREMENTS`

`S18-ACFT-R-1000` The S18 airplane shall have a means to decelerate on ground

$$\text{airplane\_decelerate\_on\_ground}$$

`S18-ACFT-R-1100` The S18 airplane shall have a means to decelerate the wheels on the ground

$$\text{airplane\_decelerate\_wheels\_on\_ground}$$

`S18-ACFT-R-1110` The S18 airplane shall have pilot-controlled wheel braking capability

$$\text{airplane\_pilot\_controlled\_braking}$$

`S18-ACFT-R-1120` The S18 airplane shall have an autobraking capability during landing and RTO

$$\text{airplane\_autobraking}$$

`S18-ACFT-R-1130` The S18 airplane shall have anti-skid braking

$$\text{airplane\_antiskid}$$

`S18-ACFT-R-1140` The S18 airplane shall provide interface for Wheel Brake System status and annunciations

$$\text{airplane\_wbs\_status\_interface}$$

`S18-ACFT-R-0181` The S18 airplane shall decelerate the wheels on gear retraction

$$\text{airplane\_decelerate\_wheels\_on\_gear\_retract}$$

`S18-ACFT-R-0182` The S18 airplane shall be capable of decelerating the wheels differentially

$$\text{airplane\_decelerate\_differentially}$$

`S18-ACFT-R-0183` The S18 airplane shall have a parking brake to prevent airplane motion when parked

$$\text{airplane\_parking\_brake}$$

`S18-ACFT-R-0184` The S18 airplane shall have hydraulically driven braking

$$\text{airplane\_hydraulic\_braking}$$

`S18-ACFT-R-0185` The S18 airplane shall have an autobrake override that can be initiated by the flight crew

$$\text{airplane\_autobrake\_override}$$

`S18-ACFT-R-0186` The S18 airplane shall meet safety requirements while operating in an average atmospheric radiation environment per the IEC 62396 standard with an altitude of 40000 feet and a latitude of 45 degrees

$$\text{airplane\_radiation\_qualified}$$

`S18-ACFT-R-0933` The Wheel Brake System decelerate the wheels on the ground function shall be developed as FDAL A

$$\text{airplane\_decelerate\_function\_fdal\_a}$$

`S18-ACFT-R-1322` No single failure or event shall result in the complete loss of wheel brake and the loss of one thrust reverser

$$\neg(\text{complete\_loss\_wheel\_brake} \land \text{loss\_one\_thrust\_reverser})$$

`S18-ACFT-R-1385` Complete loss of wheel braking shall be less than 1.0E-07 for a landing

$$\text{p\_complete\_loss\_wheel\_braking\_per\_landing} < 1 \times 10^{-7}$$

`S18-ACFT-R-1550` Loss of power from both hydraulic subsystems powered by the engines shall not lead to complete loss of wheel braking

$$(\text{both\_engine\_hyd\_failed} \Rightarrow \neg\text{complete\_loss\_wheel\_brake})$$

`S18-ACFT-R-1551` Wheel Brake System shall include an emergency accumulator to supply hydraulic power to the wheel brakes

$$\text{airplane\_has\_emergency\_accumulator}$$

`S18-ACFT-R-1552` The Alternate/Emergency Brake System hydraulic equipment and piping shall be installed aft of the engine 1 UERF trajectory envelope

$$\text{alt\_emer\_aft\_of\_uerf}$$

`S18-ACFT-R-1600` Two redundant control lanes shall be provided between the Electric Brake Unit (EBU) and each of the two Alternate/Emergency Meter Valves

$$\text{ebu\_two\_redundant\_lanes}$$

2 Axioms

$$\text{flight\_duration\_hours} = 5$$

$$\text{p\_loss\_elec\_bus\_per\_flight} = \text{lambda\_loss\_elec\_bus\_per\_hour} \cdot \text{flight\_duration\_hours}$$

$$\text{p\_loss\_pedal\_per\_flight} = \text{lambda\_loss\_pedal\_per\_hour} \cdot \text{flight\_duration\_hours}$$

3 Internal consistency

- `PSSA_REQUIREMENTS`: SAT
- `ALT_PSSA_REQUIREMENTS`: SAT
- `SPEC_REQUIREMENTS`: SAT
- `BSCU_PSSA_REQUIREMENTS`: SAT
- `BSCU_ASSUMPTIONS_TO_WBS`: SAT
- `BSCU_SPEC_REQUIREMENTS`: SAT
- `AIRPLANE_REQUIREMENTS`: SAT

4 Directional entailment

`SPEC_REQUIREMENTS` entails each `PSSA_REQUIREMENTS`

- `E12-1`: PASS
- `E12-2`: PASS
- `E12-3`: PASS
- `E13-1`: PASS
- `E13-2`: PASS
- `E13-3`: PASS
- `E13-4`: PASS
- `E13-5`: PASS
- `E13-6`: PASS
- `E13-7`: FAIL
  text: When "HYD 1 Enable" output is disabled, then "Alt/Emer Ctrl" output shall be enabled
  witness: `alt_emer_ctrl_on=False, hyd1_enable_on=False`
- `E13-8`: PASS
- `E13-9`: PASS
- `E14-1`: PASS
- `E14-2`: PASS
- `E14-3`: FAIL
  text: The probability of 7 or more wheel speed sensors erroneous or inoperative will be less than 1.0E-07 per flight
  witness: `p_seven_or_more_sensors_per_flight=1/10000000`
- `E14-4`: FAIL
  text: The probability of loss of an airplane electrical power bus will be less than 1.0E-04 per flight
  witness: `p_loss_elec_bus_per_flight=1/10000`
- `E14-5`: FAIL
  text: The probability of loss of a Left brake pedal position input will be less than 1.0E-04 per flight
  witness: `p_loss_pedal_per_flight=1/10000`
- `E14-6`: PASS
- `E14-7`: PASS

Summary: 15 passed, 4 failed.

`BSCU_SPEC_REQUIREMENTS` entails each `BSCU_PSSA_REQUIREMENTS`

- `BSCU-001`: PASS

- `BSCU-002`: PASS
- `BSCU-003`: PASS
- `BSCU-004`: PASS
- `BSCU-005`: PASS
- `BSCU-006`: PASS
- `BSCU-007`: PASS
- `BSCU-008`: PASS
- `BSCU-009`: PASS
- `BSCU-010`: PASS
- `BSCU-011`: PASS

Summary: 11 passed, 0 failed.

5 Restatement equivalence

- `E14-1` ⇔ `E28-WBS-ASMP-1`: EQUIVALENT
- `E14-2` ⇔ `E28-WBS-ASMP-2`: EQUIVALENT
- `E14-3` ⇔ `E28-WBS-ASMP-3`: EQUIVALENT
- `E14-4` ⇔ `E28-WBS-ASMP-4`: NOT EQUIVALENT
  `E14-4`: The probability of loss of an airplane electrical power bus will be less than 1.0E-04 per flight
  `E28-WBS-ASMP-4`: The failure rate for loss of an airplane electrical power bus will be less than 1.0E-04 per hour of flight
  `E14-4` entails `E28-WBS-ASMP-4`: PASS
  `E28-WBS-ASMP-4` entails `E14-4`: FAIL; witness: `p_loss_elec_bus_per_flight=1/2048`
- `E14-5` ⇔ `E28-WBS-ASMP-5`: NOT EQUIVALENT
  `E14-5`: The probability of loss of a Left brake pedal position input will be less than 1.0E-04 per flight
  `E28-WBS-ASMP-5`: The failure rate for loss of a left brake pedal position input will be less than 1.0E-04 per hour of flight
  `E14-5` entails `E28-WBS-ASMP-5`: PASS
  `E28-WBS-ASMP-5` entails `E14-5`: FAIL; witness: `p_loss_pedal_per_flight=1/2048`
- `E14-6` ⇔ `E28-WBS-ASMP-6`: EQUIVALENT
- `E14-7` ⇔ `E28-WBS-ASMP-7`: EQUIVALENT

Summary: 5 equivalent, 2 not equivalent.